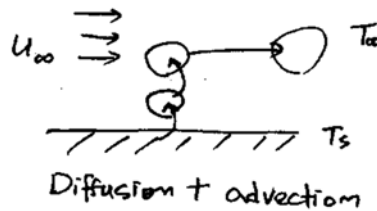


10. Introduction to heat convection



Conduction: HT in solid (diffusion by random microscopic motion)

Convection: HT with fluid flow (bulk motion) = diffusion + advection

Note)

- Diffusion (conduction) always exists, regardless whether advection occurs (there is bulk flow) or not.
- Convection $\supset$ Conduction

- Newton's law of cooling

$$q'' = h(T_s - T_\infty)$$

$$q = \int_{A_s} q'' dA_s = (T_s - T_\infty) \int_{A_s} h dA_s = \bar{h} A_s (T_s - T_\infty)$$

- On the solid surface, convection = conduction $\therefore$ No-slip BC $\mathbf{v}=0$

$$\Rightarrow q'' = h(T_s - T_\infty) = -k \left. \frac{\partial T}{\partial y} \right|_{y=0}$$

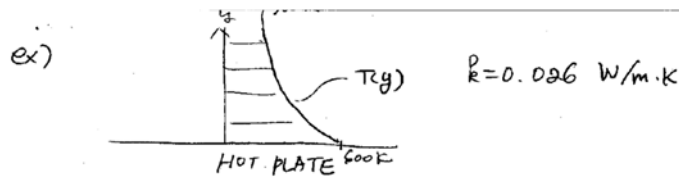
$$\Rightarrow \therefore h = \frac{-k \left. \frac{\partial T}{\partial y} \right|_{y=0}}{T_s - T_\infty} = \frac{\text{thermal conductivity} \times \text{temp. gradient on the surface}}{\text{temp. difference}}$$

<Note>

- ① Newton's law of cooling looks simple but the complexity is condensed in h
- ② Newton's law of cooling defines h

$$h \equiv \left. \frac{q''}{(T_s - T_\infty)} \right|$$

- ③ The above equation tells you how to compute h from the temperature field



$$T(y) = 300(1 + e^{-y}) \quad [K]$$

$$T(0) = T_s = 600K$$

$$T(\infty) = T_\infty = 300K$$

$$\frac{dT}{dy} = -300e^{-y}$$

Q) What is the heat transfer coefficient h ?

$$q'' = h(T_s - T_\infty) \quad h = \frac{q''}{T_s - T_\infty} = \frac{-k \frac{dT}{dy} \big|_{y=0}}{600 - 300}$$

$$= \frac{-0.026 \times (-300)}{300} = 0.026 \text{ W/m}^2\text{K}$$

- ④ $\frac{\partial T}{\partial y}$ is not linear any more but a complicated function affected by the flow field as well as by thermal conditions
- ⑤ h is not a material property but problem specific (varies with x and t) e.g.,



- ⑥ Average HT coefficient $\bar{h}$ (spatial average)

$$q = \bar{h}A(T_s - T_\infty): \text{average HT coefficient}$$

$$q'' = h(T_s - T_\infty): \text{local HT coefficient}$$

$$\bar{h} = \frac{1}{A_0} \int_{A_0} h(\mathbf{r}) dA$$

$$\text{In 1-d case, } \bar{h} = \frac{1}{l} \int_0^l h(x) dx$$

